

## TIANSHUI DAEI, China

WMO: 570140

Lat: 34.5614N

Lon: 105.8739E

Elev: 1085

StdP: 88.95

Time Zone: 8.00 (E08)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |              |            |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|--------------|------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%          |            |             | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-9.9 | (c)<br>-8.3 | (d)<br>-17.6                  | (e)<br>0.9 | (f)<br>-2.8 | (g)<br>-15.8 | (h)<br>1.1 | (i)<br>-2.1 | (j)<br>6.8            | (k)<br>-1.6 | (l)<br>6.3 | (m)<br>-0.9 | (n)<br>1.4            | (o)<br>290 | (p)<br>0.414 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>9.8             | (c)<br>32.6     | (d)<br>20.1 | (e)<br>31.2 | (f)<br>19.9 | (g)<br>29.9 | (h)<br>19.6 | (i)<br>22.6         | (j)<br>28.9 | (k)<br>21.9 | (l)<br>28.2 | (m)<br>21.3 | (n)<br>27.1 | (o)<br>3.2           | (p)<br>140 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.9                            | 17.8 | 25.3 | 20.2 | 17.0 | 24.7 | 19.5 | 16.3 | 24.4 | 73.1          | 29.6 | 70.2 | 28.3 | 67.8 | 27.1 | 25.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 6.3           | (b) 5.5 | (c) 4.8 | (e) -13.2                  | (f) 35.1 | (g) 2.0            | (h) 1.6 | (i) -14.6                                          | (j) 36.3 | (k) -15.8  | (l) 37.2 | (m) -16.9  | (n) 38.1 | (o) -18.3  | (p) 39.3 |    |    |
| (a) 6.3           | (b) 5.5 | (c) 4.8 | (e) -13.6                  | (f) 23.2 | (g) 2.0            | (h) 0.9 | (i) -15.0                                          | (j) 23.8 | (k) -16.2  | (l) 24.4 | (m) -17.3  | (n) 24.9 | (o) -18.7  | (p) 25.6 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.5  | -0.9  | 2.8   | 9.0   | 14.6  | 18.1  | 21.9  | 24.2  | 23.5  | 18.1  | 12.6  | 6.1   | 0.0  | (6)  |
| (7)  |                                                                                  | DBStd       | 9.19  | 3.11  | 3.52  | 4.12  | 3.52  | 2.87  | 2.20  | 2.31  | 2.87  | 2.77  | 3.05  | 3.36  | 2.52 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1060  | 339   | 204   | 69    | 6     | 0     | 0     | 0     | 0     | 0     | 11    | 120   | 311  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2634  | 598   | 436   | 289   | 118   | 39    | 2     | 0     | 1     | 37    | 178   | 366   | 569  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1989  | 0     | 1     | 38    | 144   | 251   | 356   | 441   | 419   | 243   | 93    | 4     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 521   | 0     | 0     | 0     | 6     | 33    | 108   | 182   | 162   | 29    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 4300  | 0     | 0     | 8     | 110   | 354   | 859   | 1529  | 1291  | 138   | 10    | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 1421  | 0     | 0     | 1     | 13    | 75    | 277   | 578   | 452   | 24    | 1     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.0   | 1.8   | 2.1   | 2.2   | 2.2   | 2.0   | 2.1   | 2.2   | 1.9   | 1.7   | 1.6   | 1.6   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 539   | 4     | 6     | 18    | 44    | 57    | 67    | 95    | 88    | 92    | 47    | 15    | 4    | (15) |
| (16) |                                                                                  | PrecMax     | 773   | 11    | 17    | 42    | 81    | 121   | 174   | 195   | 218   | 180   | 114   | 41    | 18   | (16) |
| (17) |                                                                                  | PrecMin     | 325   | 0     | 0     | 0     | 11    | 11    | 14    | 26    | 16    | 34    | 13    | 3     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 122   | 3     | 4     | 9     | 21    | 29    | 36    | 41    | 53    | 45    | 25    | 10    | 4    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 11.9  | 17.0  | 24.2  | 28.5  | 30.7  | 33.2  | 34.5  | 34.4  | 29.5  | 24.5  | 18.4  | 10.9 | (19) |
| (20) |                                                                                  |             | MCWB  | 4.2   | 7.2   | 11.4  | 14.6  | 16.2  | 18.0  | 21.2  | 21.5  | 18.1  | 15.5  | 10.7  | 4.1  | (20) |
| (21) |                                                                                  | 2%          | DB    | 8.9   | 13.7  | 21.4  | 26.2  | 28.7  | 31.4  | 32.8  | 32.1  | 26.6  | 21.7  | 16.0  | 8.7  | (21) |
| (22) |                                                                                  |             | MCWB  | 2.7   | 6.0   | 10.3  | 13.5  | 16.0  | 18.0  | 20.7  | 21.0  | 18.0  | 14.5  | 9.6   | 3.0  | (22) |
| (23) |                                                                                  | 5%          | DB    | 6.6   | 11.5  | 19.1  | 24.4  | 27.1  | 29.8  | 31.4  | 30.7  | 24.7  | 19.8  | 13.5  | 7.0  | (23) |
| (24) |                                                                                  |             | MCWB  | 1.2   | 4.8   | 9.3   | 13.0  | 15.6  | 18.0  | 20.5  | 20.9  | 17.5  | 13.6  | 7.9   | 2.1  | (24) |
| (25) |                                                                                  | 10%         | DB    | 4.6   | 9.1   | 16.6  | 22.2  | 25.4  | 28.1  | 30.0  | 29.3  | 23.1  | 18.0  | 11.7  | 5.4  | (25) |
| (26) |                                                                                  |             | MCWB  | 0.1   | 3.5   | 8.1   | 12.1  | 15.2  | 18.0  | 20.4  | 20.4  | 16.8  | 12.7  | 7.3   | 1.1  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 5.0   | 8.5   | 13.2  | 16.8  | 18.8  | 21.7  | 23.8  | 23.5  | 22.0  | 16.7  | 11.6  | 5.1  | (27) |
| (28) |                                                                                  |             | MCDWB | 10.6  | 15.8  | 20.5  | 26.4  | 26.0  | 26.9  | 29.7  | 29.0  | 24.5  | 22.5  | 17.4  | 9.7  | (28) |
| (29) |                                                                                  | 2%          | WB    | 3.0   | 6.5   | 11.2  | 15.1  | 17.8  | 20.7  | 22.7  | 22.7  | 19.6  | 15.5  | 10.0  | 3.7  | (29) |
| (30) |                                                                                  |             | MCDWB | 8.1   | 12.5  | 19.8  | 23.2  | 24.8  | 26.0  | 28.8  | 29.2  | 23.3  | 19.6  | 14.3  | 7.7  | (30) |
| (31) |                                                                                  | 5%          | WB    | 1.6   | 5.2   | 9.9   | 14.0  | 17.1  | 19.9  | 22.0  | 22.0  | 18.5  | 14.7  | 9.1   | 2.5  | (31) |
| (32) |                                                                                  |             | MCDWB | 5.9   | 10.6  | 17.6  | 21.5  | 24.0  | 26.0  | 28.3  | 28.7  | 22.0  | 18.4  | 12.4  | 6.4  | (32) |
| (33) |                                                                                  | 10%         | WB    | 0.3   | 3.9   | 8.7   | 12.9  | 16.3  | 19.0  | 21.4  | 21.4  | 17.7  | 13.7  | 8.1   | 1.4  | (33) |
| (34) |                                                                                  |             | MCDWB | 4.3   | 8.2   | 15.3  | 20.2  | 22.7  | 25.3  | 27.3  | 27.6  | 21.3  | 17.0  | 10.9  | 4.8  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 10.2  | 10.0  | 11.4  | 12.6  | 11.8  | 10.9  | 9.8   | 9.4   | 8.1   | 9.0   | 10.0  | 10.6 | (35) |
| (36) |                                                                                  |             | MCDBR | 14.4  | 15.2  | 16.4  | 16.7  | 16.2  | 15.7  | 13.4  | 11.9  | 12.0  | 12.6  | 13.8  | 13.4 | (36) |
| (37) |                                                                                  |             | MCWBR | 8.9   | 8.4   | 7.2   | 6.2   | 5.6   | 4.5   | 3.7   | 3.2   | 4.4   | 5.5   | 7.6   | 8.3  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 13.4  | 14.4  | 14.9  | 13.4  | 12.0  | 10.7  | 9.3   | 9.2   | 7.8   | 10.2  | 11.7  | 12.3 | (38) |
| (39) |                                                                                  |             | MCWBR | 8.6   | 8.3   | 7.1   | 5.7   | 4.8   | 3.8   | 3.2   | 2.9   | 3.4   | 5.1   | 6.7   | 7.9  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.481 | 0.552 | 0.599 | 0.599 | 0.575 | 0.559 | 0.561 | 0.602 | 0.562 | 0.542 | 0.484 | 0.440 | (40) |      |
| (41) |                                                                                  | taud        | 1.775 | 1.660 | 1.609 | 1.648 | 1.766 | 1.858 | 1.907 | 1.782 | 1.831 | 1.803 | 1.841 | 1.889 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 670   | 668   | 683   | 715   | 741   | 751   | 746   | 704   | 707   | 672   | 664   | 685   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 174   | 218   | 248   | 250   | 224   | 204   | 194   | 216   | 196   | 186   | 161   | 146   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.57  | 3.05  | 4.07  | 5.01  | 5.31  | 5.29  | 5.02  | 4.51  | 3.30  | 2.87  | 2.56  | 2.42  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.25  | 0.45  | 0.37  | 0.48  | 0.45  | 0.37  | 0.65  | 0.55  | 0.33  | 0.42  | 0.38  | 0.31  | (45) |      |

## Historical Trends

|                             |  | DBAvg  | Heating |       | Cooling |       |         | Degree-Days |         |         |  |
|-----------------------------|--|--------|---------|-------|---------|-------|---------|-------------|---------|---------|--|
|                             |  | 99% DB | 99% DP  | 1% DB | 1% WB   | 1% DP | HDD10.0 | HDD18.3     | CDD10.0 | CDD18.3 |  |
| (46) Station Only           |  | N/A    | N/A     | N/A   | N/A     | N/A   | N/A     | N/A         | N/A     | N/A     |  |
| (47) Regional (7 neighbors) |  | +0.24  | N/A     | N/A   | N/A     | N/A   | N/A     | -89         | +70     | N/A     |  |

Nomenclature: See separate page